

Method / seed	Scored	F_{ctx}	F_{tool}	F_{pipe}	Pending	Terminal	Total
AI-Scientist-v2 (10 labs) / 42	4	0	0	0	1	4	5
AI-Scientist-v2 (10 labs) / 43	0	0	0	0	5	0	5
AI-Scientist-v2 (10 labs) / 44	0	0	0	0	5	0	5
AI-Researcher (10 labs) / 42	0	2	2	0	1	4	5
AI-Researcher (10 labs) / 43	0	0	0	0	5	0	5
AI-Researcher (10 labs) / 44	0	0	0	0	5	0	5
Public-controller matrix	4	2	2	0	22	8	30

Table 1: Public-controller execution outcomes across five endpoints and three seeds. Terminal counts include both scored runs and verified controller failures. Pending denotes unresolved work. These counts describe the public-controller adapter protocol.

BioCoLoop: PUBLIC-HARNESS COMPARISON

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Comparison branch draft. Public frameworks use task-adapted native experimental controllers and the common biological fitting interface. The tables distinguish verified prediction scores, execution failures and pending runs. Automatic updates await visual review before mainline delivery.

Public-controller comparison status. At 2026-09-24 14:49:13 UTC, 4/30 public-controller runs have independently verified scores, 2 have verified context-limit failures, 2 required-tool failures, and 0 native-pipeline failures; 22 remain pending. Thus 8/30 runs are terminal, distinct from the scored fraction. The seed-42 pilot is matched across methods; three-seed means appear only for three verified scores. Execution outcomes describe the specified task adaptations and local-Qwen backend. Failed runs have no predictive score.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
Fixed model (1 lab)	81.76	<u>23.97</u>	<u>29.94</u>	25.56	<u>19.40</u>
Qwen direct (1 lab)	<u>88.82</u>	20.21	21.73	14.15	13.43
AI-Scientist-v2 (10 labs)	Pending	37.12	31.63	<u>22.74</u>	23.88
AI-Researcher (10 labs)	Pending	F_{ctx}	F_{tool}	F_{ctx}	F_{tool}
BioCoLoop (10 labs)	94.60	37.12	31.63	<u>22.74</u>	23.88

Table 2: Matched seed-42 comparison. Scores are multiplied by 100; bold/underline mark best/second-best values, including ties. Public controllers use task-adapted interfaces. Only complete seed sets are averaged; execution outcomes are detailed in Appendix C. F_{ctx} , F_{tool} and F_{pipe} denote context, tool and pipeline failures; Pending is unresolved.

Model / method	DTI	Proteomics	Cell perturbation		
	BindingDB AUROC $\uparrow$	PTPC AP $\uparrow$	VCC Top-1 $\uparrow$	Norman Top-1 $\uparrow$	Tahoe Top-1 $\uparrow$
A. Core comparison, seeds 42–44 (mean $\pm$ SD)					
Fixed model (1 lab)	82.83 $\pm$ 1.22	<u>25.05</u> $\pm$ 4.56	29.94 $\pm$ 0.00	25.56 $\pm$ 0.00	<u>19.40</u> $\pm$ 0.00
Qwen direct (1 lab)	89.89 $\pm$ 1.02	20.13 $\pm$ 0.07	21.09 $\pm$ 4.38	14.15 $\pm$ 0.00	15.42 $\pm$ 6.21
BioCoLoop (10 labs)	93.76 $\pm$ 1.11	36.99 $\pm$ 0.22	31.54 $\pm$ 0.08	<u>22.25</u> $\pm$ 0.45	22.39 $\pm$ 2.59
B. Matched research-controller comparison, seed 42					
Qwen direct (1 lab)	<u>88.82</u>	<u>20.21</u>	<u>21.73</u>	<u>14.15</u>	<u>13.43</u>
AI-Scientist-v2 (10 labs)	Pending	37.12	31.63	22.74	23.88
AI-Researcher / final (10 labs)	Pending	37.12	31.63	22.74	Pending
BioCoLoop (10 labs)	94.60	37.12	31.63	22.74	23.88

Table 3: Main results. A: prespecified three-seed core mean $\pm$ SD. B: matched seed-42 task-adapted controllers with 10 laboratories, 12 designs and ≤ 6 candidates. Scores are $\times 100$; bold/underline mark best/second-best within each panel; Pending is unresolved.

Method / transport	DTI AUROC	Proteomics AP	VCC Top-1	Norman Top-1	Tahoe Top-1
AI-Researcher / final adapter	Pending	37.12	31.63	22.74	Pending
BioCoLoop	94.60	37.12	31.63	22.74	23.88

Table 4: Final-adapter comparison at seed 42 with ten laboratories. AI-Researcher retains its native early-stop decision under a six-candidate maximum; no candidate is imputed to spend unused budget. All candidate fits and held-out scores use the shared scientific protocol. Values are multiplied by 100; bold marks the better value or a tie among scored pairs. Original-adapter outcomes remain in Table 2.

TASK-TRANSPORT REPAIR

This separate study removes repeated task packets from stage prompts and accepts compatible single-tool envelopes. Six candidates are a maximum, so a native early stop is retained and no additional candidate is imputed. Candidate training and held-out scorers are unchanged.